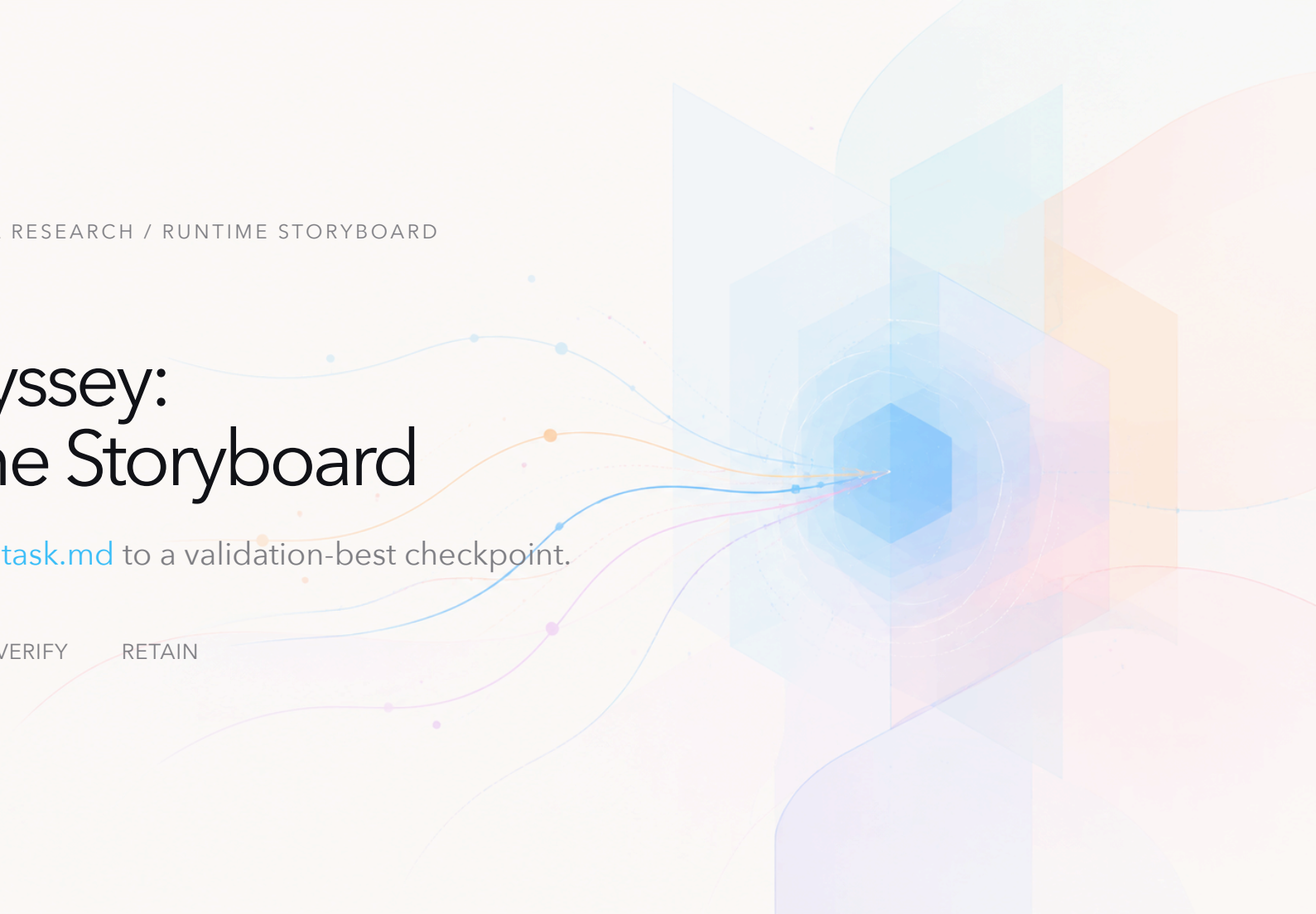


AUTONOMOUS ML RESEARCH / RUNTIME STORYBOARD

SciOdyssey: Runtime Storyboard

A live run from [task.md](#) to a validation-best checkpoint.

READ ACT VERIFY RETAIN



The run starts by reading the contract.

Before a Scientist changes code, the runtime loads the task, operator constraints, and scoring authority.

01 task.md

READY

What to solve.

Rank short videos for each user and interaction context.

KuaiRand-Pure · long_view · GAUC + nDCG@5

02 PERSONAL.md

READY

How to work.

macOS / Apple Silicon, uv, ≤ 15 min per experiment.

never print credentials

COMMAND / STEP

```
./scripts/research-agent step \  
  --cli codex --target /absolute/path/to/project --allow-edits
```

Experience enters as files, not unfinished thought.

RESEARCH WORLD / PERSISTENT

research world

- └ evidence & provenance
- └ verified knowledge
- └ current research state
- └ fallible priors
- └ State + implementation

- READ facts with evidence
- READ negative results and failures
- READ reusable implementation

A fresh Scientist inherits experience without inheriting the previous reasoning trajectory.

The Scientist turns constraints into experiments.

01

INSPECT

data · evaluator · code



02

HYPOTHESIZE

choose a scientific question



03

IMPLEMENT

modify · train · recover



04

EVALUATE

interpret the evidence

LIFECYCLE / CONTROL SURFACE

init → step → run → resume

A metric counts only when the evaluator contract says it does.

LABEL	long_view	fixed prediction label
PRIMARY	mean(GAUC, nDCG@5)	organizer evaluator
FULL	124,909 public-validation rows	hidden test is organizer-controlled
STOP	3 Full evaluations with $\Delta_{\text{primary}} \leq 0.002$	Smoke / Medium are neutral

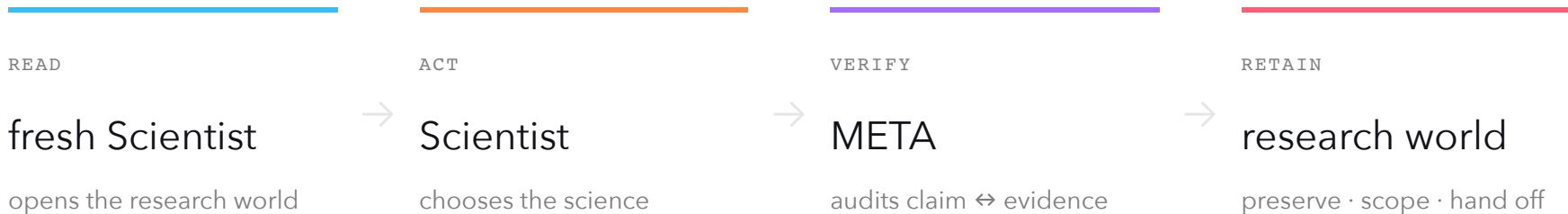
starter_kit/evaluate.py → [unchanged scoring semantics](#) → public validation only during development

The E001–E013 record turns search into an auditable run.

E001	SCREEN	target-encoding variants rejected	MEDIUM
E003	RECOVER	rich FM established a valid checkpoint	FULL
E008	ENSEMBLE	five-seed FM became the stronger baseline	FULL
E013	RETAIN	eight-seed 46-field FM selected	FULL

13 named experiments 7 Full evaluations 0 post-launch manual scientific interventions

The handoff moves evidence forward and cognition back to zero.



META owns what survives. Runtime owns deterministic mechanics. Neither replaces the Scientist's scientific judgment.

Selective persistence breaks the momentum loop.

PERSISTS ACROSS TRAJECTORIES

- metrics and provenance
- failed experiments
- original reports
- reusable implementation

CONTEXT RESET

STARTS FRESH

- an independent reasoning trajectory
- room to challenge inherited framing
- experience, not momentum

The world keeps the evidence. The Scientist gets a new chance to search.

The live run retained the validation-best checkpoint.

	GAUC	nDCG@5	PRIMARY
Official five-field FM	0.6674000	0.5357000	0.6016000
Research Agent submission	0.6728421	0.5390304	0.6059363
Δ vs reference	+0.0054421	+0.0033304	+0.0043363

E013 / FULL / SELECTED

CONVERGENCE 3 / 3

4 autonomous META-Scientist cycles out of the 50-iteration cap.

0 GPU-hours · 48,240,128 total tokens including cache-read input.

The checkpoint was selected from public validation only; the final submission CSV contains 170,588 rows and passed the unchanged Starter Kit alignment checker.

A retained result is also a runnable artifact.

REPRODUCTION / PINNED ENVIRONMENT

```
cd submission/research-agent-kuairand
uv run --python 3.12.11 --with numpy==2.5.2 \
  python system/ensemble_46.py \
    --seeds 0 1 2 3 4 5 6 7 --k 16 --lr 0.001 \
    --l2 1e-5 --full --submission final/research-agent-test.csv \
    --output final/research-agent-run.json
```

ARTIFACT CHECK

170,588 rows

Starter Kit alignment checker passed

CPU / NumPy

0 GPU-hours

Preserve the research world.

Reset the researcher.